

Topology-GRPO: A Hierarchical Group Policy Optimization Algorithm with Global Topological Calibration for Intra-Graph Inter-Group Scenarios

Lanhaijian
Hongqiao University

May 25, 2026

Abstract

This paper revises and strengthens the Topology-GRPO framework by removing intra-group value functions and critics while preserving global value calibration for topological consistency. The proposed method eliminates traditional actor-critic structures within each group and relies purely on group-wise reward normalization for advantage estimation. A global value function V_{global} is retained exclusively for cross-group credit alignment and topological validation. The algorithm maintains the original intra-graph inter-group message passing, three-component topological advantage, and inter-group policy consistency constraints while achieving cleaner objectives, more stable training, and stronger theoretical alignment with the GRPO principle. We validate the necessity of global value calibration in graph-structured decision systems and provide a theoretically sound, simplified pipeline for chip design, scheduling, and molecular generation tasks.

1 Introduction

The original Topology-GRPO framework extends GRPO to graph-structured decision systems with inter-group coupling. However, value networks within individual groups introduce unnecessary complexity and deviate from the core GRPO principle: *advantages are computed purely from relative group rewards, not learned critics.*

In intra-graph inter-group scenarios:

- Local value functions are redundant and may break credit assignment.
- Global calibration is essential for cross-group coordination.
- A lightweight global value V_{global} is sufficient for topological validation.

This paper redefines Topology-GRPO with:

1. No intra-group value functions or critics.
2. Purely reward-based group advantage normalization.
3. Global value V_{global} for topological consistency only.
4. Unified clipped policy loss with inter-group fidelity constraints.

2 Preliminaries and Problem Formulation

We model the system as a graph:

$$G = (V, E, W), \quad V = \bigcup_{g=1}^k V_g, \quad E = E_{\text{intra}} \cup E_{\text{inter}}.$$

The objective remains:

$$\max_{\pi_h, \{\pi_l^{(g)}\}} \mathbb{E} \left[\sum_{g,t} \gamma^t R^{(g)} \right] \quad \text{s.t. inter-group consistency.}$$

Key definitions:

- π_h : high-level coordination policy.
- $\pi_l^{(g)}$: low-level group policy.
- $\mathcal{N}(g)$: topological neighbor group set of g .
- W_{gh} : coupling strength between groups g and h .

3 Revised Topology-GRPO Framework

3.1 Design Principles

1. **No critic within any group.**
2. **Advantage from reward statistics only.**
3. **Global value for topological calibration.**
4. **Single policy optimization objective.**

3.2 Inter-Group Message Passing

The state of group g is enhanced via neighbor information:

$$\tilde{s}_g = \text{CONCAT} (s_g, \sigma (\text{MLP} ([s_g; m_g])) \odot m_g),$$

where

$$m_{h \rightarrow g} = \text{MLP} \left([s_g^{(i)}; \bar{s}_h; W_{gh}] \right).$$

3.3 Global Value for Topological Calibration

A global value function summarizes the global system state:

$$V_{\text{global}} = f_{\text{global}} \left(\bigcup_{g=1}^k s_g \right).$$

This value is **not a critic** and is only used to stabilize cross-group credit assignment.

3.4 Three-Component Topological Advantage

The advantage function uses **no local value functions**:

$$\begin{aligned} \hat{A}_i^g = & \alpha \cdot \frac{R_i^{\text{global}} - \bar{R}^{\text{global}}}{\sigma^{\text{global}}} \\ & + (1 - \alpha) \cdot \frac{r_i^g - \bar{r}^g}{\sigma^g} \\ & + \beta \cdot \frac{r_i^g - \bar{r}^{\mathcal{N}(g)}}{\sigma^{\mathcal{N}(g)}}. \end{aligned}$$

All terms are computed from **reward normalization**.

3.5 Inter-Group Consistency Loss

We enforce policy alignment across connected groups:

$$\mathcal{L}_{\text{cons}} = \sum_{(g,h) \in E_{\text{inter}}^{\text{group}}} W_{gh} \cdot \left\| \mathbb{E} \pi_l^{(g)} - \mathbb{E} \pi_l^{(h)} \right\|^2.$$

3.6 Final Objective Function

$$\begin{aligned} \mathcal{L}^{\text{Topo-GRPO}} = & -\mathbb{E} \left[\min \left(r_t(\theta) \hat{A}^g, \text{clip}(r_t, 1 \pm \epsilon) \hat{A}^g \right) \right] \\ & + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}} \\ & + \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}). \end{aligned}$$

4 Training Algorithm

Algorithm 1 Revised Topology-GRPO Training

- 1: Initialize high-level policy π_h , group policies $\{\pi_l^{(g)}\}$, message encoders
 - 2: Initialize global value network V_{global} for topological calibration
 - 3: **while** not converged **do**
 - 4: Construct intra-group and inter-group edges $E_{\text{intra}}, E_{\text{inter}}$
 - 5: Collect parallel trajectories across all groups
 - 6: **for** each group g **do**
 - 7: Perform inter-group message passing to obtain $\tilde{s}_g$
 - 8: Compute global, local, and neighbor rewards
 - 9: Compute three-component topological advantage $\hat{A}^g$ (no critic)
 - 10: **end for**
 - 11: Compute clipped policy loss, consistency loss, and KL divergence
 - 12: Update policies by minimizing $\mathcal{L}^{\text{Topo-GRPO}}$
 - 13: Decay λ_{cons} gradually
 - 14: **end while**
 - 15: **return** $\pi_h, \{\pi_l^{(g)}\}$
-

5 Theoretical Properties

- **GRPO-aligned:** no intra-group critic, pure reward-based advantage.
- **Topologically consistent:** global value ensures cross-group coherence.
- **Stable:** reduced variance from removing local critics.
- **Scalable:** maintains message-passing efficiency.

6 Application Scenarios

6.1 Chip Design

Groups: modules; edges: timing nets; consistency: timing closure.

6.2 Integrated Airline Scheduling

Groups: flights, crew, gates; consistency: time-window alignment.

6.3 Quantum-Ready Molecular Generation

Groups: scaffold, functional groups; consistency: chemical validity.

7 Conclusion

This revised Topology-GRPO eliminates intra-group critics and strictly adheres to GRPO principles while retaining global topological calibration. The framework is simpler, more stable, and theoretically cleaner for intra-graph inter-group decision-making.